

Week 02 — ARMA Models and the PACF

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The big picture. Three building blocks (white noise, MA, AR) combine into ARMA; the ACF and PACF are the fingerprints used to identify each.

The ARMA family is the workhorse of linear time series modelling. We start from **white noise** — the indivisible “atom” of randomness — and build **moving average** (MA), **autoregressive** (AR) and combined **ARMA** processes by feeding white noise through finite linear filters and linear recursions. Two diagnostic curves run through the whole story: the autocorrelation function (ACF, ρ_τ) and its less-obvious partner, the **partial autocorrelation function** (PACF, α_τ). The punchline, in one small table: the **ACF identifies the MA order** q , the **PACF identifies the AR order** p .

The one idea to carry through the week

Everything is built from white noise. An **MA** is white noise pushed forward through a *finite filter* (it forgets after q steps). An **AR** feeds its own past back in (its memory *never* fully dies). The ACF and PACF are just two different ways of looking at that memory — and they cut off at exactly the orders q and p .

Key definitions

Definition — White noise (purely random process)

$\{X_t\}$ is **white noise** if it is a sequence of *uncorrelated*, mean-zero, finite-variance variables: $\mathbb{E}[X_t] = 0$, $\text{Var}(X_t) = \sigma^2 < \infty$, and $\text{Cov}(X_s, X_t) = 0$ for $s \neq t$. Equivalently

$$\gamma_\tau = \begin{cases} \sigma^2 & \tau = 0, \\ 0 & \tau \neq 0, \end{cases} \quad \rho_\tau = \begin{cases} 1 & \tau = 0, \\ 0 & \tau \neq 0. \end{cases}$$

It is automatically (weakly) stationary and is the **building block** of every model below. We write ε_t for mean-zero white noise of variance σ_ε^2 .

Uncorrelated $\neq$ independent

White noise only constrains the first two moments, so it need not be i.i.d. or Gaussian. “White” refers to its *flat* spectral density (equal power at all frequencies), mirroring the flat ACF.

Definition — Moving average MA(q)

With $\{\varepsilon_t\}$ mean-zero white noise,

$$X_t = \mu - \sum_{j=0}^q \theta_j \varepsilon_{t-j}, \quad \theta_0 = -1, \theta_q \neq 0.$$

(The convention $\theta_0 = -1$ makes the leading term ε_t ; simple examples use the zero-mean form $X_t = \varepsilon_t - \theta\varepsilon_{t-1}$.) An MA(q) is just a *finite linear filter* of white noise, hence **always stationary** — the only requirement is $|\theta_j| < \infty$. Its moments are

$$\mathbb{E}[X_t] = \mu, \quad \gamma_\tau = \begin{cases} \sigma_\varepsilon^2 \sum_{j=0}^q \theta_j \theta_{j+|\tau|} & |\tau| \leq q, \\ 0 & |\tau| > q. \end{cases}$$

The **ACVS cuts off at lag** q — this is the signature used to read q off the ACF.

Definition — Autoregressive AR(p)

With $\{\varepsilon_t\}$ mean-zero white noise,

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} + \varepsilon_t, \quad \phi_p \neq 0.$$

Unlike MA, there *are* constraints on $\{\phi_j\}$ for stationarity, and there is no simple closed form for the ACVS in general.

Definition — ARMA(p, q) and causality

$\{X_t\}$ is **ARMA**(p, q) if

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} - \sum_{k=0}^q \theta_k \varepsilon_{t-k}, \quad \theta_0 = -1,$$

i.e. in backshift form ($B^j X_t = X_{t-j}$) $\Phi(B)X_t = \Theta(B)\varepsilon_t$. It is **causal** if $X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ with $\sum_j |\psi_j| < \infty$ (depends only on present and past noise; the $\{\psi_j\}$ are the MA(∞) weights). A mean-zero MA(q) is always causal.

Definition — Best linear predictor & partial correlation

The **best linear predictor** of Y from $X_1, \dots, X_n$ is $P(Y) = \sum_j \beta_j X_j$ minimising the MSE; it is characterised by the **orthogonality equations** — the error is uncorrelated with every regressor, $\text{Cov}(Y - P(Y), X_k) = 0$. The **partial correlation** of X, Y given confounders Z is the correlation of the residuals after removing the best linear effect of Z from each:

$$\rho_{XY \bullet Z} = \text{Corr}(X - P_Z(X), Y - P_Z(Y)).$$

Definition — Partial autocorrelation function (PACF)

For mean-zero stationary $\{X_t\}$, α_τ is the partial correlation of X_t and $X_{t+\tau}$ given the intervening values $Z = (X_{t+1}, \dots, X_{t+\tau-1})$:

$$\alpha_\tau = \text{Corr}(X_{t+\tau} - \hat{X}_{t+\tau}, X_t - \hat{X}_t),$$

with $\hat{X}$ the best linear predictors from Z . **Regression form:** if $P_{X_0, \dots, X_{\tau-1}}(X_\tau) = \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}$, then $\alpha_\tau = \alpha_{\tau,\tau}$, the *last* regression coefficient (and $\alpha_1 = \rho_1$).

Key results & formulas

Proposition — Causal AR(1): from recursion to MA(∞)

For $X_t = \phi X_{t-1} + \varepsilon_t$ with $|\phi| < 1$, iterating gives the causal representation and its autocovariance:

$$X_t = \sum_{k=0}^{\infty} \phi^k \varepsilon_{t-k} \quad (\psi_k = \phi^k),$$
$$\gamma_\tau = \frac{\sigma_\varepsilon^2}{1 - \phi^2} \phi^{|\tau|}, \quad \rho_\tau = \phi^{|\tau|}.$$

Idea: substitute $X_{t-1} = \phi X_{t-2} + \varepsilon_{t-1}$ repeatedly (remainder $\phi^n X_{t-n} \rightarrow 0$), then use the white-noise filter trick. *Why it matters:* the ACF $\phi^{|\tau|}$ **tails off** (never exactly 0) — which is precisely why we need the PACF.

Proposition — ACF $\leftrightarrow$ PACF (prediction equations)

For mean-zero stationary $\{X_t\}$ and $1 \leq k \leq \tau$,

$$\rho_k = \sum_{j=1}^{\tau} \alpha_{\tau,j} \rho_{k-j}.$$

Idea: split $\gamma_k = \text{Cov}(X_{\tau-k}, \hat{X}_{\tau}) + \text{Cov}(X_{\tau-k}, X_{\tau} - \hat{X}_{\tau})$; the second term is 0 by orthogonality. *Why:* these **Yule–Walker-type** equations are exactly what Durbin–Levinson solves to extract $\alpha_{\tau} = \alpha_{\tau,\tau}$.

Proposition — PACF of an AR(p) vanishes beyond lag p

For a causal AR(p), $\alpha_{\tau} = 0$ for all $\tau > p$. The *estimated* PACF at lags $> p$ is asymptotically mean 0 with variance $1/n$, giving the $\pm 1.96/\sqrt{n}$ significance bands. *Idea:* for $\tau > p$ the best predictor uses only $\phi_1, \dots, \phi_p$; setting $\alpha_{\tau,j} = \phi_j$ ($= 0$ for $j > p$) solves all prediction equations by causality, so $\alpha_{\tau,\tau} = 0$. *Why:* this is the backbone of the identification table.

Theorem — ACF & PACF signatures of ARMA

For *causal and invertible* ARMA(p, q):

	AR(p)	MA(q)	ARMA(p, q)
ACF	Tails off	Cuts off after lag q	Tails off
PACF	Cuts off after lag p	Tails off	Tails off

Read q from where the ACF cuts, p from where the PACF cuts; if neither cuts (both tail off), the model is mixed ARMA. “Tails off” = geometric / damped-sine decay with no exact zero.

Technique — Durbin–Levinson recursion (ACF $\rightarrow$ PACF)

Given the ACF, compute $\alpha_{\tau} = \alpha_{\tau,\tau}$ by

$$\alpha_{1,1} = \rho_1, \quad \alpha_{\tau,\tau} = \frac{\rho_{\tau} - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j} \rho_{\tau-j}}{1 - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j} \rho_j},$$

$$\alpha_{\tau,j} = \alpha_{\tau-1,j} - \alpha_{\tau,\tau} \alpha_{\tau-1,\tau-j}, \quad j = 1, \dots, \tau-1.$$

Works with theoretical *or* sample ACF. A clean cut to 0 at $\tau = p + 1$ flags an AR(p); the final diagonal coefficients then equal the AR coefficients.

Intuition

What the PACF actually measures

The PACF answers: “how much *new* linear information about $X_{t+\tau}$ is in X_t , once everything in between is removed?” In an AR(1), $\rho_{\tau} = \phi^{|\tau|} \neq 0$ for all τ even though X_t depends *directly* only on X_{t-1} : the lag-2 correlation is entirely *transmitted* through X_{t-1} . The PACF strips out that transmitted part, so it vanishes for $\tau > 1$ — exactly pinning the order $p = 1$.

Why MA cuts the ACF, AR cuts the PACF

An MA(q) is a window of width $q + 1$ over the noise: two observations more than q apart share *no* common noise term, so the **ACF is exactly 0 beyond lag q** . An AR(p) directly depends

on its last p values, so once those are conditioned on, an older value adds nothing: the **PACF is exactly 0 beyond lag p** . Same idea, dual diagnostics.

Why invertibility: the MA(1) twin

The MA(1) $X_t = \varepsilon_t - \theta\varepsilon_{t-1}$ and its “root-reciprocal” twin $Y_t = \eta_t - \frac{1}{\theta}\eta_{t-1}$ (with $\text{Var } \eta = \sigma_\varepsilon^2\theta^2$) have the *identical* ACVS: $\gamma_0 = \sigma_\varepsilon^2(1 + \theta^2)$, $\gamma_{\pm 1} = -\sigma_\varepsilon^2\theta$, 0 else. Second-order structure cannot tell them apart. We impose **invertibility** (root of $\Theta(z)$ outside the unit circle, i.e. $|\theta| < 1$) to choose a unique representation.

A quick word on **roots conventions** (worth memorising): *stationarity / causality* = all roots of $\Phi(z)$ outside the unit circle ($|z| > 1$); *invertibility* = all roots of $\Theta(z)$ outside. For AR(1), $\Phi(z) = 1 - \phi z$ has root $z = 1/\phi$, so $|z| > 1 \iff |\phi| < 1$. For AR(2) the conditions are $\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$, $|\phi_2| < 1$ (the “stationarity triangle”).

Durbin–Levinson identifies an AR(2)

Given $\rho_1 = \frac{2}{5}$, $\rho_2 = -\frac{1}{20}$, $\rho_3 = -\frac{1}{8}$:

$$\alpha_{1,1} = \frac{2}{5}, \quad \alpha_{2,2} = \frac{-\frac{1}{20} - \frac{2}{5} \cdot \frac{2}{5}}{1 - \frac{2}{5} \cdot \frac{2}{5}} = -\frac{1}{4}, \quad \alpha_{2,1} = \frac{2}{5} - \left(-\frac{1}{4}\right)\frac{2}{5} = \frac{1}{2},$$

$$\alpha_{3,3} = \frac{-\frac{1}{8} - \frac{1}{2}\left(-\frac{1}{20}\right) - \left(-\frac{1}{4}\right)\frac{2}{5}}{1 - \frac{1}{2} \cdot \frac{2}{5} - \left(-\frac{1}{4}\right)\left(-\frac{1}{20}\right)} = 0.$$

The PACF cuts to 0 at lag 3 $\Rightarrow$ **AR(2)**: indeed $Y_t = \frac{1}{2}Y_{t-1} - \frac{1}{4}Y_{t-2} + \varepsilon_t$ reproduces these values, and $(\alpha_{2,1}, \alpha_{2,2}) = (\frac{1}{2}, -\frac{1}{4}) = (\phi_1, \phi_2)$ recovers the AR coefficients.

Key takeaways

Key takeaways — the week’s reflexes

- **Build everything from white noise:** MA = finite filter (always stationary); AR = feedback (needs root conditions); ARMA = both.
- **The identification table is the goal:** ACF cuts after $q \Rightarrow$ MA(q); PACF cuts after $p \Rightarrow$ AR(p); both tail off $\Rightarrow$ mixed ARMA; nothing significant $\Rightarrow$ white noise. “Significant” = outside $\pm 1.96/\sqrt{n}$.
- **MA covariance reflex:** $\gamma_\tau = \sigma_\varepsilon^2 \sum_j \theta_j \theta_{j+|\tau|}$ — keep only matching noise indices; it is 0 for $|\tau| > q$.
- **Roots outside the unit circle:** $\Phi(z)$ for causality, $\Theta(z)$ for invertibility. For AR(1) this is just $|\phi| < 1$.
- **Be able to run Durbin–Levinson** from a given ACF to get $\alpha_{\tau,\tau}$, and to spot the cut-off lag.
- **First check stationarity:** an ACF decaying almost linearly (very slowly) signals non-stationarity — difference before identifying.